

## 2. Agile Frameworks & Project Practices

- Agile is a Project management and software development approach that focuses on Iterative development, Customer Collaboration, ~~requirements~~ Adapting to changing requirements, Continuous improvement.
- Instead of delivering a complete Product at the end, Agile delivers working Software in ~~small~~ small increments frequently.

### \* Scrum Framework:

- Scrum is the most Popular Agile framework used to manage Complex Projects through short development cycles called sprints.
- It's characteristics are it's iterative and incremental development, Continuous feedback, Team Collaboration, frequent Product releases.

### ① \* Kanban Basics:

- Kanban is a visual workflow management method used to improve efficiency, transparency and the Continuous delivery of work.
- It originated from Toyota's manufacturing Practices and later became Popular in Software development.

### \* Core Principles of Kanban:

- 1) Visualize work → Task are represented as cards on a Kanban board.



→ This visibility helps the team understand the Project status instantly.

2) Limit work in Progress (WIP) → The team sets a maximum number of tasks allowed in each stage.

→ The benefits of this is it prevents multitasking, Improves quality, increases focus.

3) Manage Flow → The objective is smooth movement of task from start to completion.

→ The matrix includes cycle time, lead time and throughput.

4) Make Policies explicit → In this team clearly defines the definition of Done, coding standards, Review Procedures, Testing requirements and much more.

## ②\* User stories →

→ A user story is a short description of a feature written from the end user's perspective.

## ③\* Story Points →

→ Story points measure the relative effort, complexity, uncertainty and risk involved in implementing a user story.



## \* Release Planning →

- Release Planning will determine which feature will deliver, in which version, during which time period with what resources by aligning the business goals with development activity.
- The main objective is deliver maximum customer value, priorities high impact features, estimate time to lines and allocate the resources efficiently.

## \* Velocity →

- Velocity is the number of story points completed by a team in one sprint.
- The objective of velocity is to measure over multiple sprints, maintain stable teams, focuses on value delivered rather than the numerical targets.

## \* Retrospective →

- A sprint retrospective is a meeting after each sprint where a team reflects on what went well, what went wrong, what can be improved while the main goal is continuous improvement.